Kaggle Connect X — Week 4 — Speaker Script

Kaggle Connect X 第4周 讲稿

Left side: English. Right side: Chinese. The English text is the same as the speaker notes in the PowerPoint file. Main slides 1–12: about 1240 words. At a slow pace (110 words per minute) this is about 11 minutes. Slides 13–14 are backup slides for questions. All numbers come from `presentation/numbers.json`. "Slide 5" here means the 5th slide in the file. The number printed on that slide is 4, because the title slide is counted as 0.	左边是英文，右边是中文。 英文内容和 PPT 演讲者备注里的英文相同。 正文第1–12页约 1240 个英文单词，按较慢的语速（每分钟 110 词）大约 11 分钟。 第13–14页是付録，用来回答提问。 所有数字都来自 `presentation/numbers.json`。 这里的“第5页”指文件里的第5张幻灯片。那张幻灯片上印的页码是4，因为模板把封面算作第0页。

Five questions to prepare / 重点准备的五个问题

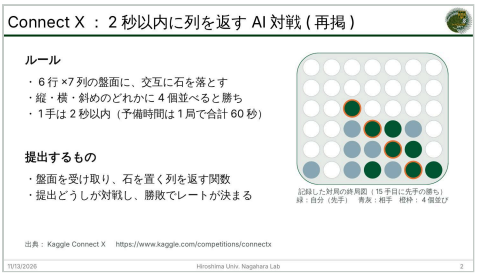
Why not just use an open-source solver? → slide 5	为什么不直接用开源求解器？→ 第5页
Why must the second player lose? → slides 3 and 8	后手为什么必输？→ 第3页、第8页
Does the "luring" strategy really work? → slides 10 and 11	“诱敌”策略真的有效吗？→ 第10页、第11页
Are 168 games enough? → slides 7 and 9	168 局样本够吗？→ 第7页、第9页
How is the ranking calculated? → slides 2 and 13	排名怎么算？→ 第2页、第13页



Slide 1 / 第1页 · about 30 s

タイトル

Script (English)	讲稿（中文）
Hello everyone. I am Yu from Team C. This is our week 4 report on Kaggle Connect X. Today I will talk about three things. First, our new agent, version 7. It is a perfect solver. Second, what we learned from 168 real games on Kaggle. Third, a small improvement, version 8, and how I tested it.	大家好，我是C组的余。这是我们 Kaggle Connect X 第4周的报告。今天讲三件事。第一，新的智能体 v7，它是一个完美求解器。第二，从 Kaggle 上 168 局真实对局里学到了什么。第三，一个小改进 v8，以及我是怎么验证它的。



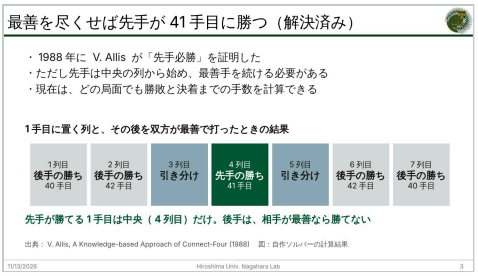
Slide 2 / 第2页（number on the slide / 幻灯片上的页码：1） · about 45 s

Connect X：2秒以内に列を返すAI対戦(再掲)

Script (English)	讲稿（中文）
First, a short review of the game. The board has 6 rows and 7 columns. Two players drop stones in turn. The first player who gets four in a line wins. The line can be vertical, horizontal or diagonal. Our agent is a function. It gets the board and returns one column. It must answer within 2 seconds. The picture on the right is a real game from our records. Green is our agent. It won at move 15.	先简单回顾一下规则。棋盘是6行7列，两个人轮流落子。先把四个子连成一线的人获胜，横、竖、斜都可以。我们的智能体是一个函数，输入棋盘，返回一个列号。每一步必须在2秒内回答。右边的图是记录里的一局真实对局，绿色是我们的智能体，它在第15手获胜。

Expected questions / 预计的提问

Q1. What happens if the agent is too slow?	问1：如果智能体算得太慢会怎样？
A. Each move has 2 seconds. There are also 60 seconds of extra time for the whole game. If the extra time is used up, the agent loses. In our 168 games we never used the extra time.	答：每一步有2秒。整局另外还有60秒的备用时间。备用时间用完就判负。我们的 168 局里，一次都没有用到备用时间。
Q2. How is the ranking calculated?	问2：排名是怎么算的？
A. By a skill rating called mu. A new submission starts at 600. A win raises mu and a loss lowers it. The leaderboard shows the mu of each team's best submission. I have a backup slide about this.	答：按一个叫 μ 的技能分来排。新提交从600分开始，赢了加分，输了减分。排行榜显示每个队最好的那次提交的 μ。我在付录里准备了一页说明。



Slide 3 / 第3页（number on the slide / 幻灯片上的页码：2） · about 55 s

最善を尽くせば先手が41手目に勝つ（解決済み）

Script (English)	讲稿（中文）
Connect Four is a solved game. In 1988, Victor Allis proved that the first player can always win. But the first player must start in the center column, and must play the best move every time. The figure shows the result of each first move. I computed it with my own solver. Only the center column wins. The win comes at move 41. The two columns next to the center give a draw. The other four columns lose. So the second player cannot win if the first player is perfect. Please remember this. It is important for today's talk.	四子棋是一个已经被解开的游戏。1988年，Victor Allis 证明了先手必胜。但先手必须从中间一列开始，并且每一步都走最优着法。这张图是第一手下在各列时的结果，是我用自己的求解器算出来的。只有中间一列能赢，在第41手获胜。中间两旁的两列是和棋，其余四列是先手输。所以，只要先手完美，后手就赢不了。请记住这一点，它对今天的内容很重要。

Expected questions / 预计的提问

Q1. Why must the second player lose?	问1：为什么后手一定会输？
A. It is a property of the game, not of our agent. With perfect play from both sides, the first player wins at move 41. Near the end the board is almost full, and one player is forced to play a bad square. This is called zugzwang. The first player can control it if it starts in the center. Our data agrees: in our 78 losses, the opponent played the best move every time.	答：这是游戏本身的性质，不是我们智能体的问题。双方都完美时，先手在第41手获胜。棋盘快满的时候，总有一方被迫下在对自己不利的格子上，这叫 zugzwang（迫移）。先手从中间开始，就能控制这件事。我们的数据也一致：输掉的 78 局里，对手每一步都是最优着法。
Q2. Is your solver's result the same as the known result?	问2：你的求解器算出的结果和已知结论一致吗？
A. Yes. The values of the seven first moves are the same as the known values. I also tested 5,020 positions with known answers. There was no mismatch.	答：一致。七个第一手的结果和已知结论相同。另外我用 5,020 个有标准答案的局面做了测试，没有一个不一致。



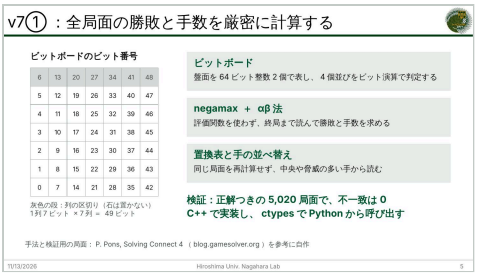
Slide 4 / 第4页（number on the slide / 幻灯片上的页码：3）· about 45 s

v7から、評価値ではなく読み切りで手を選ぶ

Script (English)	讲稿（中文）
This table shows our versions. Versions 1 to 6 choose a move with search and an evaluation function. They look some moves ahead and then estimate. Version 7 is different. It reads the game to the end. So it knows the true result of every position. Version 8 is version 7 plus one new rule. I will explain it later. Version 7 was submitted to Kaggle, and we have 168 game records from it. Version 8 is also submitted, but I have not collected its games yet.	这张表是我们各个版本的变化。v1 到 v6 用搜索加评估函数来选着法，先往前看几步，再估计局面好坏。v7 不一样，它一直算到终局，所以知道每个局面真正的结果。v8 是在 v7 上加了一条新规则，后面会讲。v7 已经提交到 Kaggle，我们有它的 168 局对局记录。v8 也提交了，但它的对局记录还没有收集。

Expected questions / 预计的提问

Q1. Is version 7 stronger than version 6?	问1：v7 比 v6 强吗？
A. In theory version 7 cannot make a mistake, so it should not be weaker. But I have not run version 7 against version 6 directly. The Kaggle scores are on the result slide.	答：理论上 v7 不会走错，所以不会更弱。但我没有让 v7 和 v6 直接对战过。Kaggle 分数在结果那一页。



Slide 5 / 第5页（number on the slide / 幻灯片上的页码：4） · about 65 s

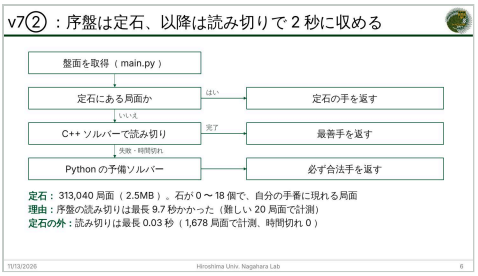
v7①:全局面の勝敗と手数を厳密に計算する

Script (English)	讲稿（中文）
This is the solver. It has three parts. First, a bitboard. The board is stored in two 64-bit integers. We can check four in a line with bit shifts. It is very fast. Second, negamax with alpha-beta pruning. We do not use an evaluation function. We search to the end of the game. So we get the exact result and the number of moves. Third, a transposition table and move ordering. We do not solve the same position twice, and we try good moves first. The method follows the public tutorial by Pascal Pons. The code is written in C++ for this project. I tested it on 5,020 positions with known answers. All answers matched.	这是求解器，由三部分组成。第一，位棋盘。整个棋盘用两个64位整数表示，用移位运算就能判断四连，非常快。第二，negamax 加 alpha-beta 剪枝。我们不用评估函数，一直搜索到终局，所以能得到精确的胜负和手数。第三，置换表和着法排序。同一个局面不重复计算，并且先试好的着法。方法参考了 Pascal Pons 的公开教程，代码是为这个项目用 C++ 写的。我用 5,020 个有标准答案的局面测试，全部一致。

Expected questions / 预计的提问

Q1. Why not just use an open-source solver?	问1：为什么不直接用开源的求解器？
A. The method is not new. I followed the public method by Pascal Pons. But we wrote our own code for four reasons. One: Kaggle runs a Python file, so we need a small library that Python can call. Two: we need a time limit and a backup, so the agent never fails. Three: for version 8 we must change how the solver chooses between equal moves. Four: the goal of this project is to learn how to build an agent.	答：方法并不新，我参考的是 Pascal Pons 公开的方法。但我们自己写代码有四个原因。一，Kaggle 运行的是 Python 文件，所以需要一个 Python 能调用的小型库。二，需要时间限制和兜底，保证智能体不会出错。三，v8 要改变求解器在同分着法之间的选择方式，必须能改代码。四，这个项目的目的就是学习怎么做智能体。
Q2. Did you write all the code yourself?	问2：代码都是你自己写的吗？
A. The code was written with an AI coding tool, Claude Code. My part was to set the goal, submit to Kaggle, collect the game records and check the results. The solver was checked with public test positions.	答：代码是用 AI 编程工具 Claude Code 写的。我负责定目标、提交 Kaggle、收集对局记录和检查结果。求解器用公开的测试局面验证过。
Q3. What does the exact score mean?	问3：精确分数是什么意思？
A. Positive means the player to move wins. A bigger number means a faster win. Zero is a draw.	答：正数表示轮到走的一方能赢，数字越大赢得越快。零是和棋，负数是输。

Negative means a loss.	



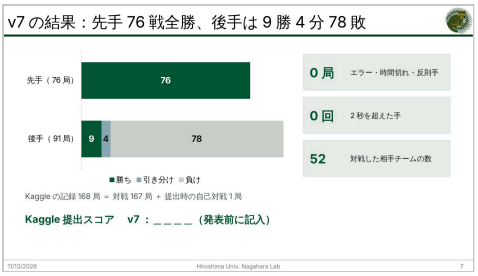
Slide 6 / 第6页（number on the slide / 幻灯片上的页码：5） · about 60 s

v7②:序盤は定石、以降は読み切りで2秒に収める

Script (English)	讲稿（中文）
The solver is exact, but the opening is slow. Some hard opening positions took up to 9.7 seconds. The limit is 2 seconds. So I computed the opening in advance. This is the opening book. It has about 313 thousand positions, with 0 to 18 stones on the board. The flow is on the left. First, look in the book. If the position is there, return the book move. If not, call the C++ solver. After the book, the solver is fast. The slowest position took 0.03 seconds in my test. If the C++ library fails, a pure Python solver is used. The agent always returns a legal move.	求解器是精确的，但开局阶段算得慢。一些难的开局局面最长要 9.7 秒，而限制是2秒。所以我把开局提前算好，这就是开局库。它有大约 31 万个局面，覆盖棋盘上有 0 到 18 个子的局面。流程在左边。先查开局库，查到就返回库里的着法。查不到就调用 C++ 求解器。出了开局库以后求解器很快，我测到最慢的局面是 0.03 秒。如果 C++ 库出问题，就用纯 Python 的求解器。智能体一定会返回合法的着法。

Expected questions / 预计的提问

Q1. How did you build the opening book?	问1：开局库是怎么生成的？
A. I start from the empty board. On our turn I store only the best move. On the opponent's turn I try every move. So the book covers every reply of the opponent. It took about half an hour on 4 CPU cores.	答：从空棋盘开始。轮到我们时只存最优的一步，轮到对手时把每一步都展开。这样对手的任何应着都在库里。在4核 CPU 上大约算了半小时。
Q2. Is a compiled library allowed on Kaggle?	问2：Kaggle 允许提交编译好的库吗？
A. The rules do not say it clearly. The submission passed the validation game and played 167 games without any error. If the library cannot be loaded, the Python solver works as a backup.	答：规则没有明确写。这次提交通过了验证局，并且打了 167 局没有任何报错。如果库加载不了，会用 Python 求解器兜底。



Slide 7 / 第7页（number on the slide / 幻灯片上的页码：6） · about 40 s

v7の結果：先手76戦全勝、後手は9勝4分78敗

Script (English)	讲稿（中文）
This is the result of version 7 on Kaggle. We have 168 game records. One is the self-play check, so there are 167 real games. As the first player we played 76 games and won all of them. As the second player we played 91 games. We won 9, drew 4 and lost 78. There were no errors and no timeouts. We played against 52 teams. The Kaggle score is written at the bottom.	这是 v7 在 Kaggle 上的结果。我们有 168 局记录，其中1局是提交时的自我对弈检查，所以真实对局是 167 局。先手 76 局，全部获胜。后手 91 局，9 胜 4 平 78 负。没有报错，也没有超时。一共遇到 52 个队伍。Kaggle 分数写在最下面。

Expected questions / 预计的提问

Q1. Are 168 games enough?	问1：只有 168 局，样本够吗？
A. It depends on the claim. For 'the agent has no errors and no timeouts', it is good evidence, but not a proof. For 'the first player always wins', I do not rely on these games. The solver is exact and tested. For 'how often opponents make mistakes', it is a small sample. Only 13 games had a mistake. So I do not make strong claims about that.	答：要看是哪个结论。对“智能体不报错、不超时”，这是不错的证据，但不是证明。对“先手必胜”，我不依赖这些对局，求解器本身是精确的，并且测试过。对“对手多久犯一次错”，样本很小，只有 13 局出现失误，所以这方面我不下强结论。
Q2. Why did a perfect solver lose so many games?	问2：完美求解器为什么还输这么多？
A. All losses were as the second player. The next slide explains why they could not be avoided.	答：输的全是后手局。下一页解释为什么这些局避免不了。



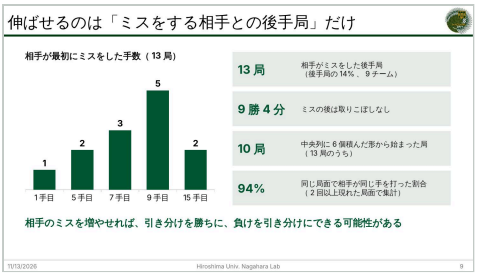
Slide 8 / 第8页（number on the slide / 幻灯片上的页码：7） · about 45 s

後手の78敗は、全手が最善の相手で防げない

Script (English)	讲稿（中文）
Why did we lose 78 games? I checked every position of every game with the solver. The result is clear. All 78 losses ended at move 41. This is the length of a perfect game. In these games the opponents made 1,638 moves. Every one was the best move. Our moves were also all best moves. So these opponents are perfect solvers too. When a perfect player moves first, the second player always loses. We cannot change these games.	为什么输了 78 局？我用求解器检查了每一局的每一个局面，结果很清楚。78 局输棋全部在第 41 手结束，这正是双方完美对弈时的长度。这些局里对手一共走了 1,638 步，每一步都是最优着法。我们的每一步也都是最优着法。所以这些对手也是完美求解器。完美的一方先走，后手一定输。这些局我们改变不了。

Expected questions / 预计的提问

Q1. Could a different move by us change the result?	问1：我们换一种走法能改变结果吗？
A. No. In these positions every move loses against perfect play. We can only choose how slowly we lose. We already lose as slowly as possible, at move 41.	答：不能。这些局面里，面对完美的对手，任何一步都是输。我们只能选择输得多慢，而现在已经是最慢的了，在第 41 手。
Q2. How do you know these opponents are solvers and not just lucky?	问2：怎么知道这些对手是求解器，而不只是运气好？
A. 1,638 moves and not one was worse than the best move. These games were against 32 different opponents. Luck cannot explain that.	答：1,638 步里没有一步比最优着法差，而且来自 32 个不同的对手，运气解释不了。



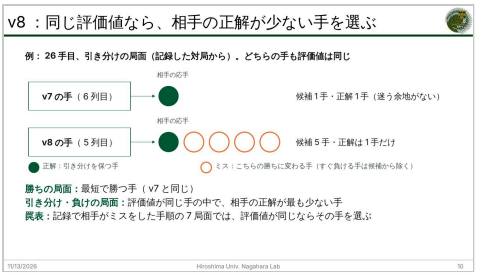
Slide 9 / 第9页（number on the slide / 幻灯片上的页码：8） · about 55 s

伸ばせるのは「ミスをする相手との後手局」だけ

Script (English)	讲稿（中文）
So where can we improve? Only in second-player games against opponents who make mistakes. There were 13 such games. That is 14 percent of our second-player games. After the mistake we did not give the advantage back: 9 wins and 4 draws. The chart shows when the first mistake came. 10 of the 13 were between move 5 and move 9. 10 of the 13 games started with six stones in the center column. One more point. Opponents are very repeatable. In the same position, they played the same move 94 percent of the time. So if a line worked once, it may work again.	那么哪里还能提升？只有“对手会犯错的后手局”。这样的对局有 13 局，占后手局的 14%。对手失误以后，我们没有把优势还回去：9 胜 4 平。图里是对手第一次失误出现在第几手，13 局里有 10 局在第5到第9手之间。13 局里有 10 局，开头都是中间一列连下6个子。还有一点：对手的走法很固定。相同的局面下，他们有 94% 的情况走同一步。所以一条路线成功过一次，就可能再成功。

Expected questions / 预计的提问

Q1. Could the 4 draws have been wins?	问1：那 4 局平局本来能赢吗？
A. In those games the opponent's mistake changed a win into a draw, not into a loss. After that, the opponent defended correctly. So the true value was a draw. We can only win if the opponent makes one more mistake. Version 8 tries to make that more likely.	答：这几局里，对手的失误只是把胜局变成了和棋，没有变成输棋。之后对手一直防守正确，所以局面的真实价值就是和棋。只有对手再犯一次错我们才能赢。v8 就是想让这种情况更容易发生。
Q2. Is 13 games enough to say anything?	问2：13 局能说明什么吗？
A. It is a small number. It only shows where to look. This is why I tested version 8 with simulated opponents, and why I say the real effect is still unknown.	答：数量很少，只能告诉我们该往哪里看。所以我用模拟对手测试了 v8，并且明确说实战效果还不知道。



Slide 10 / 第10页（number on the slide / 幻灯片上的页码：9） · about 80 s

v8：同じ評価値なら、相手の正解が少ない手を選ぶ

Script (English)	讲稿（中文）
This is the idea of version 8. Often several moves have the same exact value. Version 7 just takes the first one. Version 8 asks: which move is harder for the opponent? Here is a real example from our records. It is move 26 and the position is a draw. After the version 7 move, the opponent has only 1 possible reply. It is forced, and it is correct. After the version 8 move, the opponent has 5 possible replies. Only 1 keeps the draw. The other 4 give us a win. Both moves are equally safe for us. But the second one gives the opponent more chances to go wrong. There is also a small trap table with 7 positions. They come from recorded games where the opponent made a mistake later. The rule never picks a worse move. If we can win, we still win as fast as possible.	这是 v8 的想法。经常有好几步的精确分数是一样的。v7 直接取第一个。v8 会问：哪一步对对手来说更难？这是记录里的一个真实例子，第 26 手，局面是和棋。v7 的着法之后，对手只有 1 个可走的应着，是被迫的，而且是正确的。v8 的着法之后，对手有 5 个可走的应着，只有 1 个能保住和棋，其余 4 个都会让我们赢。两步对我们来说同样安全，但第二步给了对手更多走错的机会。另外还有一个小的陷阱表，共 7 个局面，来自对手后来犯过错的那些对局。这条规则不会选更差的着法。能赢的时候，仍然选最快赢的。

Expected questions / 预计的提问

Q1. Does this 'luring' strategy really work?	问1：这种“诱敌”策略真的有效吗？
A. My honest answer: partly, and only in simulation so far. Against a perfect opponent it changes nothing. Against simulated imperfect opponents, it helped in the positions where version 7 and version 8 differ. In full games I saw no clear difference, because such positions are rare. On Kaggle I have no data yet.	答：老实说：部分有效，而且目前只在模拟里。对完美的对手，它什么都改变不了。对模拟的不完美对手，在 v7 和 v8 选着不同的那些局面上有帮助。在完整对局里我没看到明显差别，因为这种局面很少出现。Kaggle 实战还没有数据。
Q2. Is version 8 more risky than version 7?	问2：v8 会不会比 v7 更冒险？
A. No. Version 8 only chooses among moves with the same exact score. I played version 8 against version 7 for 200 games. Each side won all its first-player games.	答：不会。v8 只在精确分数相同的着法里挑。我让 v8 和 v7 对战了 200 局，双方执先时都全胜。
Q3. What is a 'possible reply'?	问3：“可走的应着”指什么？
A. A reply that does not lose at once. Any agent can block an immediate four. So I do not count	答：指不会马上输掉的应着。任何智能体都会堵住马上要成的四连，所以这种被迫的堵，我不算作真

such forced blocks as real chances for a mistake.	正的犯错机会。



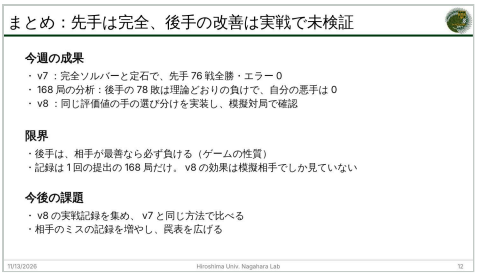
Slide 11 / 第11页（number on the slide / 幻灯片上的页码：10） · about 90 s

v8はv7に劣らないが、全体の勝率差は見えない

Script (English)	讲稿（中文）
How did I test version 8? First, version 8 against version 7, 200 games. Each side won all its first-player games. So version 8 is not weaker. Second, against the random and Negamax agents, both versions won every game. Third, I made a simulated opponent. It is a perfect solver, but 30 percent of its moves come from a weaker search. As the second player, over 1,000 games, version 7 won 828 and version 8 won 828. No difference. Fourth, I started games from the 8 positions where the two versions choose different moves. There version 8 was better. Wins went from 7 to 64, and losses from 800 to 628. But look at the details: 3 positions improved, 4 were the same, and 1 was slightly worse. So my conclusion is careful. Version 8 is not worse than version 7. It helps in some special positions. But these positions are rare, and the opponent is simulated. The effect on Kaggle is not confirmed.	我是怎么测试 v8 的？第一，v8 对 v7，200 局。双方执先时都全胜，所以 v8 没有变弱。第二，对 random 和 Negamax 智能体，两个版本都是全胜。第三，我做了一个模拟对手：它是完美求解器，但有30%的着法改用较弱的搜索来走。执后 1,000 局，v7 赢 828 局，v8 赢 828 局，没有差别。第四，我从两个版本选着不同的 8 个局面出发开始对局。在这里 v8 更好：胜局从 7 变成 64，负局从 800 变成 628。但要看细节：3 个局面变好，4 个一样，1 个略差。所以我的结论比较谨慎：v8 不比 v7 差，在一些特殊局面上有帮助，但这些局面很少，而且对手是模拟的。在 Kaggle 上的效果还没有确认。

Expected questions / 预计的提问

Q1. Why is there no difference in the 1,000 games?	问1：为什么 1,000 局里看不出差别？
A. The two versions choose the same move in almost all positions. In the recorded second-player games they differ in only 12 of about 1,805 positions. So a full game rarely reaches a position where they differ.	答：两个版本在几乎所有局面上选的着法都一样。在记录的后手对局里，约 1,805 个局面中只有 12 个不同。所以完整对局很少会走到两者不同的局面。
Q2. Is the simulated opponent realistic?	问2：模拟对手和真实对手像吗？
A. I am not sure. Real opponents made their mistakes early, mostly between move 5 and move 9. My simulated opponent can make a mistake at any move. This is a limitation of the test.	答：我不确定。真实对手的失误出现得早，大多在第5到第9手。我的模拟对手在任何一手都可能失误。这是这个测试的局限。



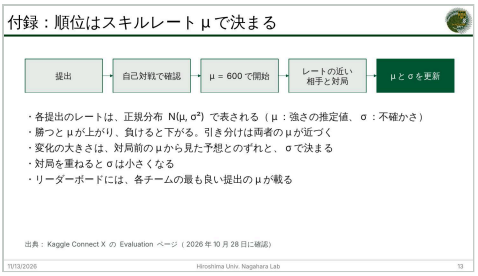
Slide 12 / 第12页（number on the slide / 幻灯片上的页码：11） · about 65 s

まとめ：先手は完全、後手の改善は実戦で未検証

Script (English)	讲稿（中文）
To sum up. Version 7 is a perfect solver with an opening book. As the first player it won all 76 games, with no errors. From 168 games we learned that our 78 losses could not be avoided. Version 8 adds a rule for equal moves. It works in simulation. There are limits. The second player loses against a perfect opponent. This is the game itself. We only have 168 games from one submission. And version 8 is not tested on real games yet. Next, I will collect real games of version 8 and compare them with version 7 in the same way. I also want to collect more opponent mistakes and extend the trap table. That is all. Thank you.	总结一下。v7 是带开局库的完美求解器。先手 76 局全胜，没有报错。从 168 局里我们知道，输掉的 78 局是避免不了的。v8 增加了一条处理同分着法的规则，在模拟里有效。局限有这些：面对完美的对手，后手一定输，这是游戏本身决定的。我们只有一次提交的 168 局。v8 还没有在实战中验证。接下来，我会收集 v8 的实战对局，用同样的方法和 v7 比较。我还想收集更多对手的失误，扩充陷阱表。以上，谢谢大家。

Expected questions / 预计的提问

Q1. Can the Kaggle score still go up?	问1：Kaggle 分数还能再涨吗？
A. As the first player, no. We already win every game. As the second player, only against opponents who make mistakes. So the possible gain is small.	答：先手不能，已经全胜了。后手只有面对会犯错的对手才有可能。所以能涨的空间很小。
Q2. What about reinforcement learning?	问2：强化学习呢？
A. For the standard board, the exact solver already plays the best move. So reinforcement learning cannot play better. It may be useful to model how opponents make mistakes. I have not tried it.	答：对标准棋盘来说，精确求解器已经走的是最优着法，强化学习不可能下得更好。它也许可以用来学习对手怎么犯错。这个我还没试过。



Slide 13 / 第13页（number on the slide / 幻灯片上的页码：12） · about 40 s

付録：順位はスキルレート μ で決まる

Script (English)	讲稿（中文）
This slide is for questions about the ranking. Each submission has a rating. It is a normal distribution with mean mu and uncertainty sigma. A new submission first plays against itself, to check that it works. Then it starts at mu 600. It plays against submissions with similar ratings. A win raises mu, a loss lowers it, and a draw moves the two values closer. The leaderboard shows the mu of each team's best submission.	这一页是为排名方面的提问准备的。每个提交有一个评分，用均值 μ 和不确定度 σ 的正态分布表示。新提交先自己和自己下一局，确认能正常运行，然后从 μ=600 开始。它会和评分接近的提交对局。赢了 μ 上升，输了下降，平局则两边的值互相靠近。排行榜显示每个队最好的那次提交的 μ。

Expected questions / 预计的问

Q1. Why is a perfect agent not far above the others?	问1：完美的智能体为什么没有远远领先？
A. This is my own reading. About half of the games are as the second player. The top agents are also perfect. Against them we win as the first player and lose as the second. So the ratings of perfect agents stay close to each other.	答：这是我自己的理解。大约一半的对局是后手。排在前面的智能体也是完美的，和它们下，先手赢、后手输。所以完美智能体之间的评分会很接近。



Slide 14 / 第14页（number on the slide / 幻灯片上的页码：13） · about 35 s

付録：よく当たる相手は、同じ局面で同じ手を打つ

Script (English)	讲稿（中文）
This table shows the opponents we met most often. For each one it shows our results as the first and as the second player. The last column shows how repeatable the opponent is. For example, the first opponent had 77 repeated positions and played the same move in 72 of them. Most of these opponents beat us every time when they move first. They are perfect solvers.	这张表是我们遇到次数最多的对手。每个对手列出了我们执先和执后的成绩。最后一列是对手走法的固定程度。比如第一个对手有 77 个重复出现的局面，其中 72 个每次都走同一步。这些对手大多数在执先时每次都赢我们，它们是完美求解器。